

# Boost Performance, Reliability, and Scalability for Your Applications

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 [systemdesignschool.io/fundamentals/load-balancer](https://systemdesignschool.io/fundamentals/load-balancer)



## Load Balancer

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Now that we know why we want to have stateless servers, let's take a look at how to do it exactly.

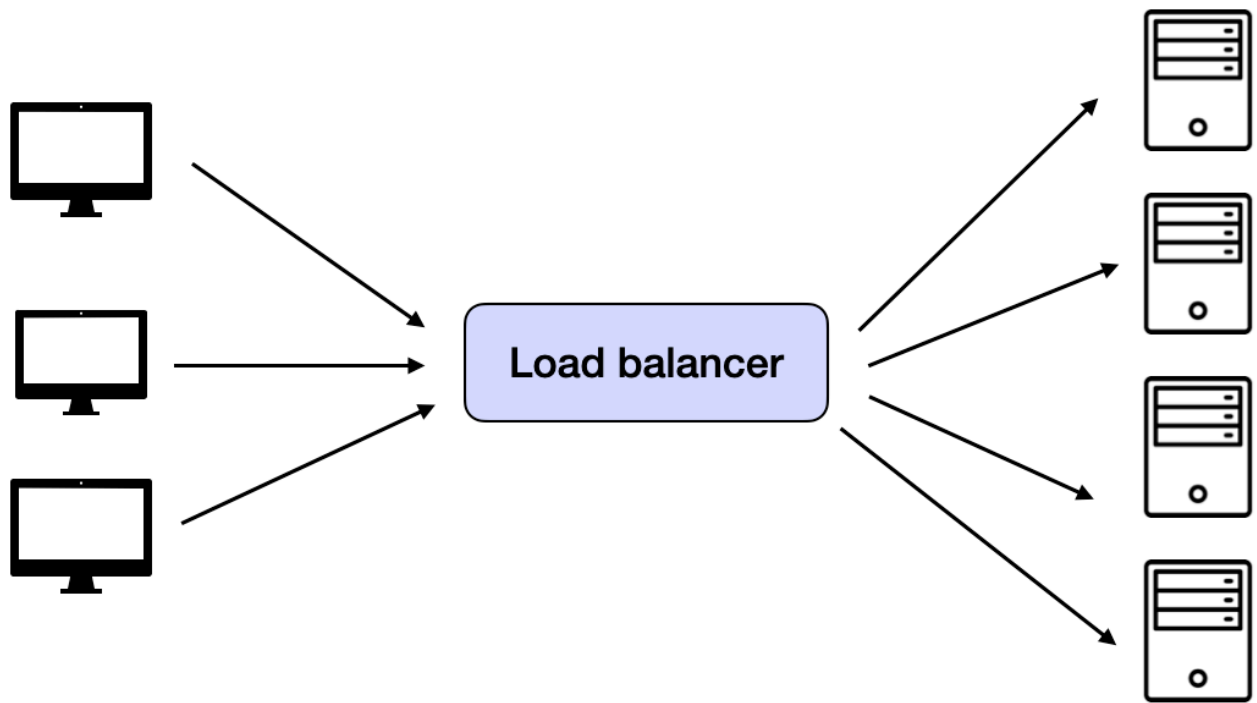
At a high level, we need two things:

- A load balancer to put in front of the servers that do the actual work to forward requests and coordinate which server is responsible for which request.
- A mechanism to launch more servers as traffic is high and wind down servers when traffic is low. The mechanism also helps in launching new servers when servers fail.

## Load balancer

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Load balancers are devices or software that act as a reverse proxy, distributing client requests across multiple servers based on predetermined algorithms. This distribution process aims to optimize resource utilization, minimize response time, and avoid overloading a single server.



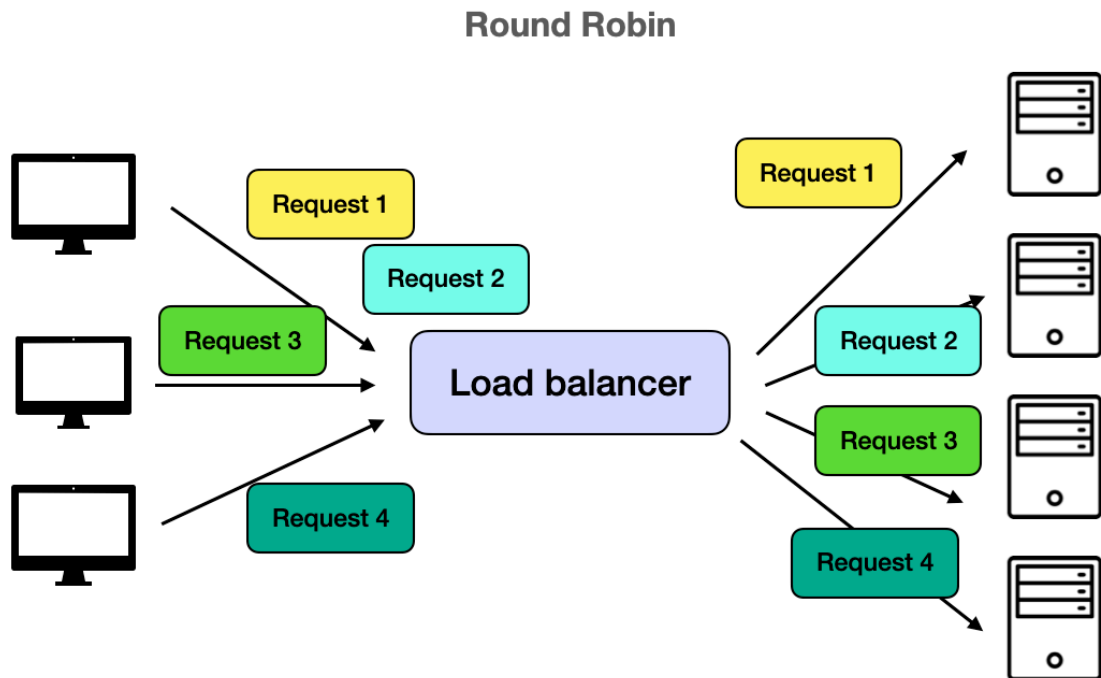
Load balancer

## Load Balancing Algorithms

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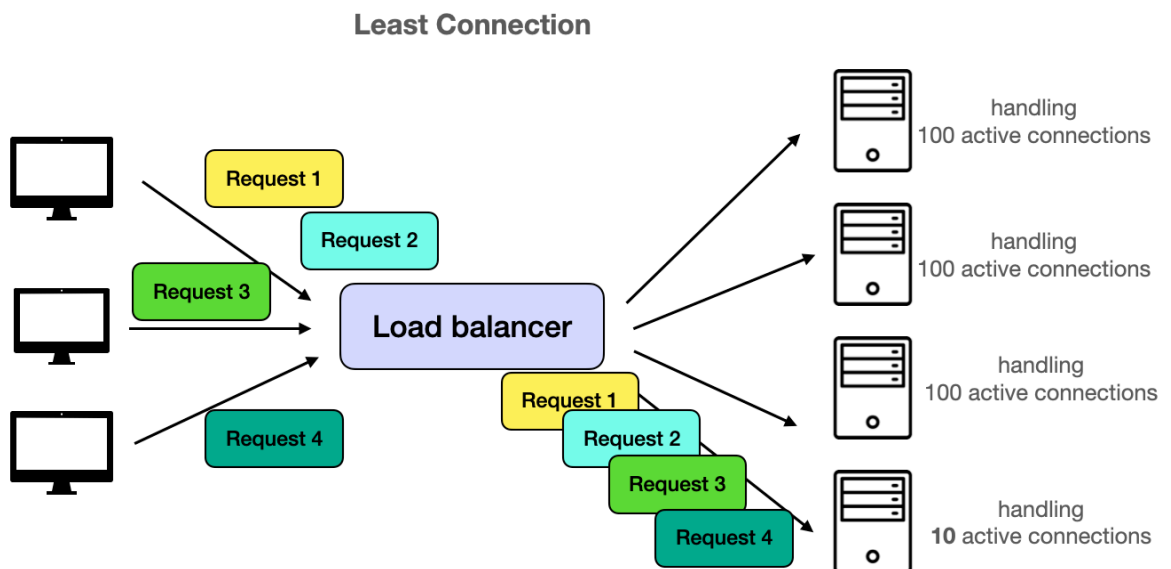
There are various load balancing algorithms, including:

1. Round Robin: Requests are distributed sequentially to available servers.



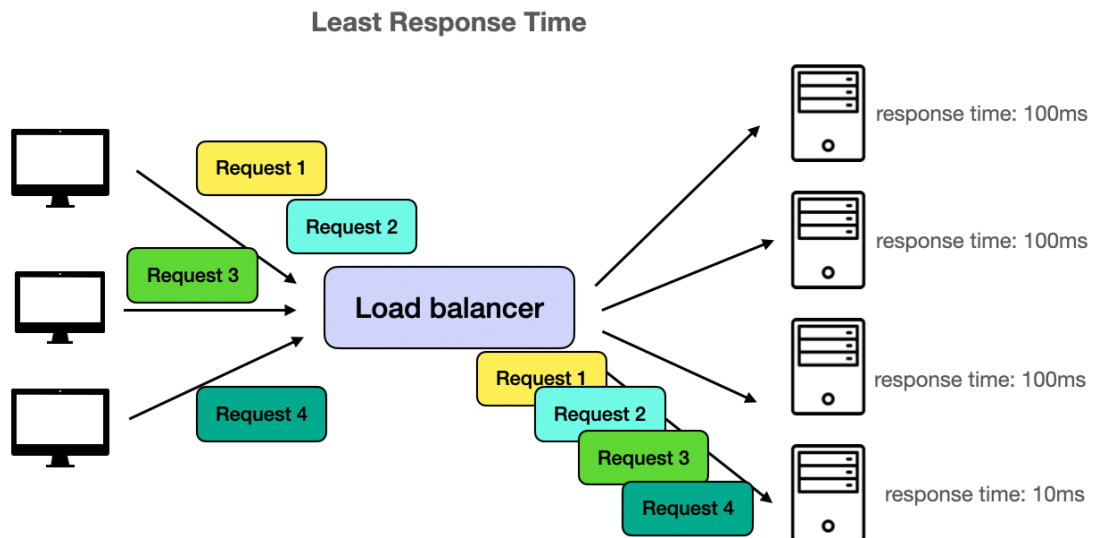
Load balancer round robin

2. Least Connections: Requests are routed to the server with the fewest active connections.



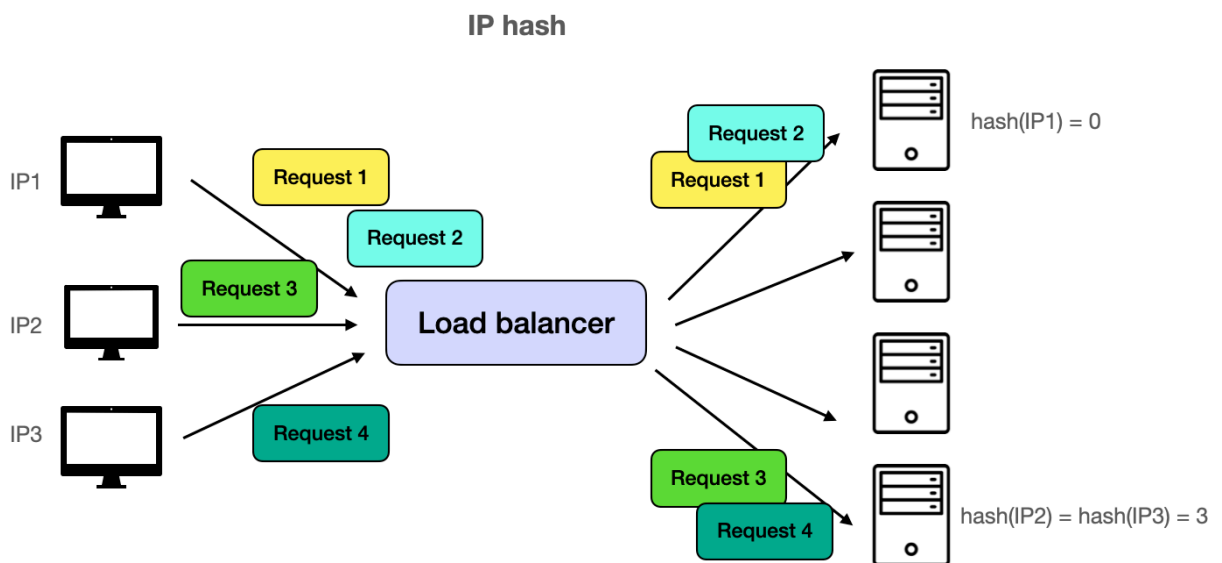
Load balancer least connection

3. Least Response Time: Requests are directed to the server with the lowest response time.



Load balancer least response time

4. IP Hash: Requests are distributed based on the client's IP address, ensuring that a specific client is always routed to the same server.



Load balancer IP Hash

On top of these, there are also

- Weighted Round Robin: Like Round Robin, except server loads are based on their compute capacity.

- URL Hash: Like IP hash, except the URL in the request is used. This is useful when we want to optimize each server for a set of URLs. For example, cache may be stored on the servers themselves and we can maximize cache hit rate using URL hash load balancing.

Load balancers play a critical role in ensuring high availability, fault tolerance, and scalability in large-scale applications. They can be implemented as hardware appliances, virtual appliances, or software solutions. Next, let's take a look at the most popular software load balancer, Nginx.

## NGINX

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NGINX is a powerful, open-source web server, reverse proxy server, and load balancer that has gained popularity due to its performance, reliability, and flexibility. It can handle a large number of concurrent connections while maintaining low memory and CPU usage, making it a popular choice for high-traffic websites and applications.

In the next section, we will build a small demo using Nginx to demonstrate how load balancing works.